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Set – 1

Question 1: (A)

Answer:

Concept of Artificial Intelligence and Its Applications

Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think, learn, and make decisions. AI systems can perform tasks that typically require human intelligence, such as understanding language, recognizing patterns, solving problems, and adapting to new situations.

AI is broadly categorized into **narrow AI** (designed for specific tasks like voice recognition) and **general AI** (capable of performing any intellectual task a human can do). With the integration of machine learning, deep learning, and natural language processing, AI has become a transformative force across industries.

Real-World Applications of AI:

- **Healthcare:** AI is used in diagnostics, such as analyzing X-rays or predicting diseases using patient data. IBM Watson, for instance, assists doctors in cancer treatment planning.

- **Finance:** AI algorithms detect fraudulent transactions and automate trading decisions based on market trends.
- **Retail:** Personalized recommendations on platforms like Amazon are powered by AI-based recommendation engines.
- **Transportation:** Self-driving cars use AI to interpret sensor data and make real-time driving decisions.
- **Customer Service:** Chatbots and virtual assistants like Siri and Alexa use AI to understand and respond to user queries.

AI continues to evolve, offering smarter automation, better decision-making, and enhanced user experiences across sectors.

(B)

Answer:

Working of a Problem-Solving Agent in AI

A **Problem-Solving Agent** is an intelligent system that decides what actions to take to achieve a specific goal. It operates in a well-defined environment and follows a structured approach to reach the desired outcome.

The working of a problem-solving agent involves the following steps:

1. **Formulate the Goal:** Define what the agent is trying to achieve. For example, a GPS navigation system aims to find the shortest route to a destination.
2. **Formulate the Problem:** Convert the goal into a problem statement. This includes defining the initial state, possible actions, transition model, and goal test.
3. **Search for a Solution:** The agent explores possible sequences of actions using search algorithms (like BFS, DFS, or A*). It evaluates different paths to find the most efficient one.
4. **Execute the Solution:** Once a solution is found, the agent performs the actions in sequence to reach the goal.

Real-Life Example:

Consider a delivery robot in a warehouse. Its goal is to pick up a package from one location and deliver it to another. It formulates the goal, maps the warehouse layout, uses a pathfinding algorithm to avoid obstacles, and executes the optimal path to complete the task.

In essence, a problem-solving agent mimics human decision-making by breaking down tasks into logical steps and using algorithms to find the best course of action.

Question 2(A):

Answer:

Search Strategies in Artificial Intelligence

Search strategies are fundamental to AI problem-solving. They help agents explore possible solutions in a structured way. These strategies are broadly classified into **uninformed (blind)** and **informed (heuristic)** search techniques.

Uninformed Search Strategies: These strategies do not use any domain-specific knowledge. They rely solely on the structure of the problem.

- **Breadth-First Search (BFS):** Explores all nodes at the current depth before moving deeper. It guarantees the shortest path but consumes more memory.
- **Depth-First Search (DFS):** Explores as deep as possible along each branch before backtracking. It uses less memory but may get stuck in infinite paths.
- **Uniform Cost Search:** Expands the least-cost node first. It's optimal but slower in large graphs.

Informed Search Strategies: These use heuristics—estimates of how close a state is to the goal—to guide the search.

- **Greedy Best-First Search:** Selects the node that appears closest to the goal based on heuristic value.
- **A Search*:** Combines path cost and heuristic to find the most efficient path. It's both complete and optimal if the heuristic is admissible.

Comparison: Uninformed search is simple but inefficient for complex problems. Informed search is faster and more accurate due to heuristic guidance. For example, solving a maze with BFS may take longer than A*, which uses distance estimates to reach the exit faster.

(B)

Answer:

Heuristic Search Techniques and Their Advantages

Heuristic search techniques enhance AI performance by using domain-specific knowledge to guide the search process. A **heuristic** is a rule or estimate that helps prioritize which paths to explore, reducing computation time and improving efficiency.

Advantages:

- **Speed:** Heuristics narrow down the search space, making problem-solving faster.
- **Efficiency:** They reduce memory usage by avoiding irrelevant paths.
- **Scalability:** Heuristic methods handle large and complex problems better than blind searches.

Common Heuristic Methods:

1. A Search*:
 - Combines actual cost (g) and estimated cost to goal (h).

- Formula: $f(n) = g(n) + h(n)$
- Example: In GPS navigation, A* finds the shortest route by considering both distance traveled and estimated distance remaining.

2. Greedy Best-First Search:

- Selects the node with the lowest heuristic value.
- Focuses on reaching the goal quickly, not necessarily optimally.
- Example: In puzzle solving (like 8-puzzle), it chooses moves that seem to bring the puzzle closer to the solution.

In summary, heuristic search techniques make AI systems smarter and more practical by guiding them toward promising solutions efficiently.

Question 3(A):

Answer:

Working of AO Algorithm and Handling AND/OR Nodes*

The **AO*** (And-Or Star) algorithm is a heuristic search method used in problem-solving where the solution involves both **AND** and **OR** relationships between subproblems. Unlike traditional search algorithms that follow a single path, AO* explores a graph structure where nodes may require multiple child nodes to be solved together (**AND**) or just one among many (**OR**).

Working Mechanism:

- The algorithm starts at the root node and expands it based on heuristic values.
- It evaluates paths using a cost function that includes both actual and estimated costs.
- If a node has **OR** branches, AO* selects the child with the lowest cost.
- If a node has **AND** branches, it must solve all connected sub-nodes together, and the total cost is the sum of their individual costs.

Handling AND vs OR Nodes:

- **OR Node:** Represents alternative solutions. AO* chooses the most promising one.
- **AND Node:** Represents a composite solution. AO* must solve all subparts to complete the parent node.

This dual handling makes AO* ideal for problems like planning, diagnosis, and multi-step reasoning, where some tasks depend on multiple conditions being met simultaneously.

(B)

Answer:

Expert Systems: Architecture, Functioning, and Applications

An **Expert System** is an AI-based software that mimics the decision-making ability of a human expert. It uses a knowledge base and inference engine to solve complex problems in specific domains like medicine, engineering, or finance.

Architecture:

1. **Knowledge Base:** Stores domain-specific facts and rules.
2. **Inference Engine:** Applies logical rules to the knowledge base to derive conclusions.
3. **User Interface:** Allows users to interact with the system.
4. **Explanation Module:** Justifies the reasoning behind decisions.

Working Mechanism:

- The user inputs a query or problem.
- The inference engine matches the input against rules in the knowledge base.
- It uses reasoning techniques like forward or backward chaining to reach conclusions.
- The system provides recommendations or diagnoses based on its findings.

Advantages:

- Consistent decision-making
- Availability 24/7
- Reduces human error
- Can handle large volumes of data

Real-World Applications:

- **Medical Diagnosis:** Systems like MYCIN assist doctors in identifying diseases.
- **Customer Support:** Automated troubleshooting guides.
- **Financial Analysis:** Risk assessment and investment advice.
- **Agriculture:** Crop disease identification and treatment suggestions.

Expert systems are valuable tools in domains requiring specialized knowledge and logical reasoning.

Question 4(A):

Answer:

Role of AI in Game Playing and Minimax Algorithm

Artificial Intelligence plays a significant role in game playing by enabling machines to make strategic decisions, simulate human-like behavior, and adapt to dynamic environments. AI-powered game agents can analyze game states, predict opponent moves, and optimize their own strategies to win.

In games like chess, tic-tac-toe, or Go, AI uses search algorithms and evaluation functions to explore possible moves and outcomes. It can simulate future game states and choose the most promising path to victory.

One of the most widely used algorithms in game playing is the **Minimax algorithm**. It is a decision-making strategy used in two-player, zero-sum games where one player's gain is another's loss.

How Minimax Works:

- The algorithm builds a game tree of possible moves.
- Each node represents a game state, and branches represent player actions.
- The **maximizing player** tries to maximize the score, while the **minimizing player** tries to minimize it.
- The algorithm recursively evaluates all possible outcomes and selects the move that leads to the best guaranteed result.

Example:

In tic-tac-toe, the AI evaluates all possible moves and chooses the one that either leads to a win or blocks the opponent from winning. Minimax ensures that even in worst-case scenarios, the AI makes the most rational move.

AI in game playing enhances user experience, enables competitive bots, and contributes to research in strategic planning and decision-making.

(B)

Answer:

Knowledge Representation in AI and Core Methods

Knowledge Representation (KR) is a fundamental aspect of Artificial Intelligence that deals with how information is structured and stored so that machines can understand, reason, and make decisions. It bridges the gap between raw data and intelligent behavior.

Importance of KR:

- Enables AI systems to simulate human reasoning.
- Facilitates problem-solving and learning.
- Supports communication between agents and users.

- Helps in building expert systems and intelligent assistants.

Core Methods of Knowledge Representation:

1. Semantic Networks:

- Represent knowledge as a graph of nodes (concepts) and edges (relationships).
- Example: “Cat is a mammal” can be represented as a link between “Cat” and “Mammal”.

2. Frames:

- Structured templates that hold attributes and values for objects or situations.
- Example: A “Car” frame may include slots for brand, model, and engine type.

3. Production Rules:

- Use IF-THEN statements to encode knowledge.
- Example: IF temperature > 100 THEN alert = “Overheat”.

4. Logic-Based Representation:

- Uses formal logic (propositional or predicate) to express facts and rules.
- Example: $\forall x (\text{Human}(x) \rightarrow \text{Mortal}(x))$ means all humans are mortal.

These methods allow AI systems to reason, infer new knowledge, and interact meaningfully with their environment.

Question 5(A):

Answer:

Reasoning in AI and Forward Chaining Steps

In Artificial Intelligence, **reasoning** refers to the process by which machines draw conclusions from available knowledge. It enables AI systems to make decisions, solve problems, and infer new facts based on existing data and rules. Reasoning can be deductive, inductive, or abductive, depending on how conclusions are derived.

Forward chaining is a data-driven reasoning technique used in rule-based systems. It starts with known facts and applies inference rules to derive new conclusions until a goal is reached.

Steps in Forward Chaining:

1. **Initialize with Known Facts:** The system begins with a set of input facts.
2. **Match Rules:** It scans the rule base to find rules where the conditions match the known facts.
3. **Apply Rule:** When a rule is matched, its conclusion is added to the fact base.

4. **Repeat:** The process continues iteratively, applying new rules as more facts are added.
5. **Goal Check:** The system stops when the desired goal or conclusion is reached.

Example:

If the system knows “It is raining” and has a rule “IF it is raining THEN carry an umbrella,” it will infer “carry an umbrella” using forward chaining.

This method is widely used in expert systems and diagnostic tools where decisions are built step-by-step from available data.

(B)

Answer:

AI Planning Systems and Approaches to Goal Achievement

AI Planning Systems are intelligent frameworks that enable machines to formulate a sequence of actions to achieve specific goals. These systems operate in dynamic environments and must consider constraints, resources, and possible outcomes to make effective decisions.

Planning in AI involves:

- Defining the initial state and goal state.
- Identifying available actions and their effects.
- Generating a plan that transitions from the initial to the goal state.

Two Common Planning Approaches:

1. State-Space Planning:

- Focuses on exploring all possible states and transitions.
- Uses search algorithms like BFS or DFS to find a path from the initial state to the goal.
- Example: A robot navigating a grid to reach a target location.

2. Hierarchical Task Network (HTN) Planning:

- Breaks down complex goals into smaller, manageable tasks.
- Uses predefined methods to decompose tasks recursively.
- Example: In logistics, delivering a package may be broken into tasks like loading, routing, and unloading.

AI planning systems are used in robotics, logistics, game AI, and automated scheduling. They help machines act intelligently and adaptively to achieve goals efficiently.

Question 6(A):

Answer:

Probabilistic Reasoning and Bayes' Theorem in AI

Probabilistic reasoning is a method in Artificial Intelligence that deals with uncertainty. Instead of relying on fixed rules, it uses probability theory to make decisions based on incomplete or noisy data. This approach is essential in real-world scenarios where outcomes are not always deterministic.

One of the key tools in probabilistic reasoning is **Bayes' Theorem**, which helps update the probability of a hypothesis based on new evidence.

Bayes' Theorem Formula:

$$P(H|E) = \frac{P(E|H) \cdot P(H)}{P(E)}$$

Where:

- $P(H|E)$: Posterior probability (probability of hypothesis H given evidence E)
- $P(H)$: Prior probability (initial belief about H)
- $P(E|H)$: Likelihood (probability of evidence E given H)
- $P(E)$: Total probability of evidence

Real-World Example:

Consider a spam filter in an email system.

- Hypothesis (H): Email is spam
- Evidence (E): Email contains the word "win"
Using Bayes' theorem, the system calculates the probability that the email is spam given the presence of the word "win," based on past data. If the posterior probability is high, the email is marked as spam.

Bayesian reasoning allows AI systems to make informed decisions even when data is uncertain or incomplete.

(B)

Answer:

Comparing Supervised, Unsupervised, and Reinforcement Learning

AI systems learn from data using different learning paradigms. The three main types are **Supervised**, **Unsupervised**, and **Reinforcement Learning**, each suited to different tasks.

1. Supervised Learning

- Involves labeled data (input-output pairs).

- The model learns to map inputs to correct outputs.
- Example: Email classification (spam vs non-spam).
- Application: Fraud detection, medical diagnosis, sentiment analysis.

2. Unsupervised Learning

- Uses unlabeled data.
- The model identifies patterns or groupings without predefined labels.
- Example: Customer segmentation based on shopping behavior.
- Application: Market research, anomaly detection, recommendation systems.

3. Reinforcement Learning

- The agent learns by interacting with an environment and receiving feedback (rewards or penalties).
- It aims to maximize cumulative reward over time.
- Example: A robot learning to walk by trial and error.
- Application: Game AI, robotics, self-driving cars.

Comparison:

- Supervised learning is accurate but requires labeled data.
- Unsupervised learning is flexible but harder to evaluate.
- Reinforcement learning is dynamic and ideal for sequential decision-making.

Each method contributes uniquely to building intelligent systems capable of learning and adapting.